Date: _____

ASSIGNMENT

Subject
Programming Fundamental

Submitted By:
NooR Asma

Submitted To:
Sir ASad Majeed

Section:
BSCS (D)

GHAZI UNIVERSITY DGKHAN

Hi-Time

Task 1

Program 1:

```cpp
#include <iostream>
using namespace std;
int main(){
    char letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
return 0;
}
```

```
A

[Program finished]
```

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Task:-1

Type and save the following Program in visual studio. Run these programs and observe their output

Program 1

```
#include <iostream>
using namespace std;
    int main()
    {
        char letter;
        letter = 'A';
      Cout << letter << "In";
      letter = 'B';
      Cout << letter << "In";
        Cin >> letter;
          return 0;
    }
```

Out Put:

A
B

observation: In this program letter is a char type and we stored "A" and 'B' in it. So it printed A and B normally. This is basic use of char.

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Program 2:

```cpp
#include <iostream>
using namespace std;
int main(){
    char letter;
letter = 65; // 65 is ASCII code of 'A'
cout << letter << '\n';
letter = 66; // 66 is ASCII code of 'B'
return 0;
}

```

```
A

[Program finished]
```

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Program :- 2

```
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 65;
cout << letter << "\n";
    letter = 66;
cout << letter << "\n";
    cin >> letter;
    return 0;
```

output

A
B

In this program letter is still char but we stored numbers 65 and 66. output was still A and B. This prove that computer stores characters as ASCII numbers. 65 mean 'A' and 66 means 'B'.

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Program 3:

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4      int letter;
5  letter = 'A';
6  cout << letter << '\n';
7  letter = 'B';
8  return 0;
9  }
```

```
65

[Program finished]
```

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Program 3:

```
#include <iostream>
using namespace std;
int main()
{
    int letter;
letter = 'A';
cout << letter << "\n";
  letter = 'B';
cout << letter << "\n";
  cin >> letter;
  return 0;
}
```

output

```
65
66
```

Observation:-

letter is int type and we stored 'A' and 'B'. output was 65 and 66. It show that when we store a character in int variable. It show its ASCII value, not the character itself.

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Task 2

```cpp
#include <iostream>
using namespace std;
int main(){
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Sum = " << a+b << endl;
    cout << "Subtraction = " << a-b << endl;
    cout << "Multiplication = " << a*b << endl;

    cout << "Division = " << a/b << endl;
    cout << "Remainder = " << a%b << endl;

    return 0;
}
```

```
Enter two integers:
```

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Task :- 2

Take two integers as input from the user apply arithematic appaaratus on them.

(+, -, *, /) as point them as Screen-

Program :-

```
#include <iostream>
using namespace std;
  int main()
{
    int num1, num2;
  Cout << "Enter two integers";
    Cin >> num1 >> num2;
  Cout << "Addition" << num1 + num2
  << endl;
  Cout << "Multiplication" "num1 num2
  << endl;
  Cout << "Division" << num1 / num2
  << endl;
      return 0;
}
```

Out Put:

If user enter two integers : 20, 5

Addition : 25 Multiplication : 100

Substraction : 15 Division : 4

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Task 3

```cpp
#include <iostream>
using namespace std;
int main(){
cout << "Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;

cout << val1 << "\n" << val2 << "\n" << val3
<< "\n";
cout << val3 << "\n" << val2 << "\n" << val1
<< "\n";
return 0;
}
```

```
Please enter your 3 numbers:10 20 30
10
20
30
30
20
10
```

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Tasks : 3

Write output of following code.

```
#include <iostream>
using namespace std;
int main()
{
    int val 1;
    int val 2;
    int val 3;
    Cout << " please enter your 3 number ";
    Cin >> val 1 >> val 2 >> val 3;
    Cout << val 1 << "In" << val 2 <<
"In" << val 3 << "in";
    Cout << val 3 << "in" << val2 <<
"In" << val 1 << "in";
    system ("PAUSE");
    return 0;
}
```

output

If you enter → 10 20 30

Please enter your 3 numbers: 10 20 30

10
20
30
30
20
10

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Task 4

```cpp
#include<iostream>
#include<cmath> // for sqrt and pow
using namespace std;
int main()
{
    double x1, y1, x2, y2, d;
    cout << "Enter x1 y1 x2 y2: ";
    cin >> x1 >> y1 >> x2 >> y2;

    d = sqrt( pow(x2-x1, 2) + pow(y2-y1, 2) );
    cout << "Distance = " << d;

    return 0;
}
```

```
Enter x1 y1 x2 y2:
```

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Task 4:- write C++ program to calculate the distance between two point, Note X_1, Y_1, X_2, Y_2 are all double value

```
#include <iostream>
using namespace std;
int main()
{
double x1, y1, x2, y2;
double distance;
cout<< "Enter x1 and y1:";
cin >> x1 >> y1;
cout<< "Enter x2 and y2:";
cin>> x2 >> y2;
d= sqrt((x2-x1)*(x1-x1)+
(y2-y1)*(y1-y1));
cout<< "Distance b/w two point is
<< distance << end;
return;
}
```

output: Suppose two point (1,2), (4,6)

Enter X_1 and Y_1 : 1, 2

Enter X_2 and Y_2 : 4 6

Distance between two points is : 5

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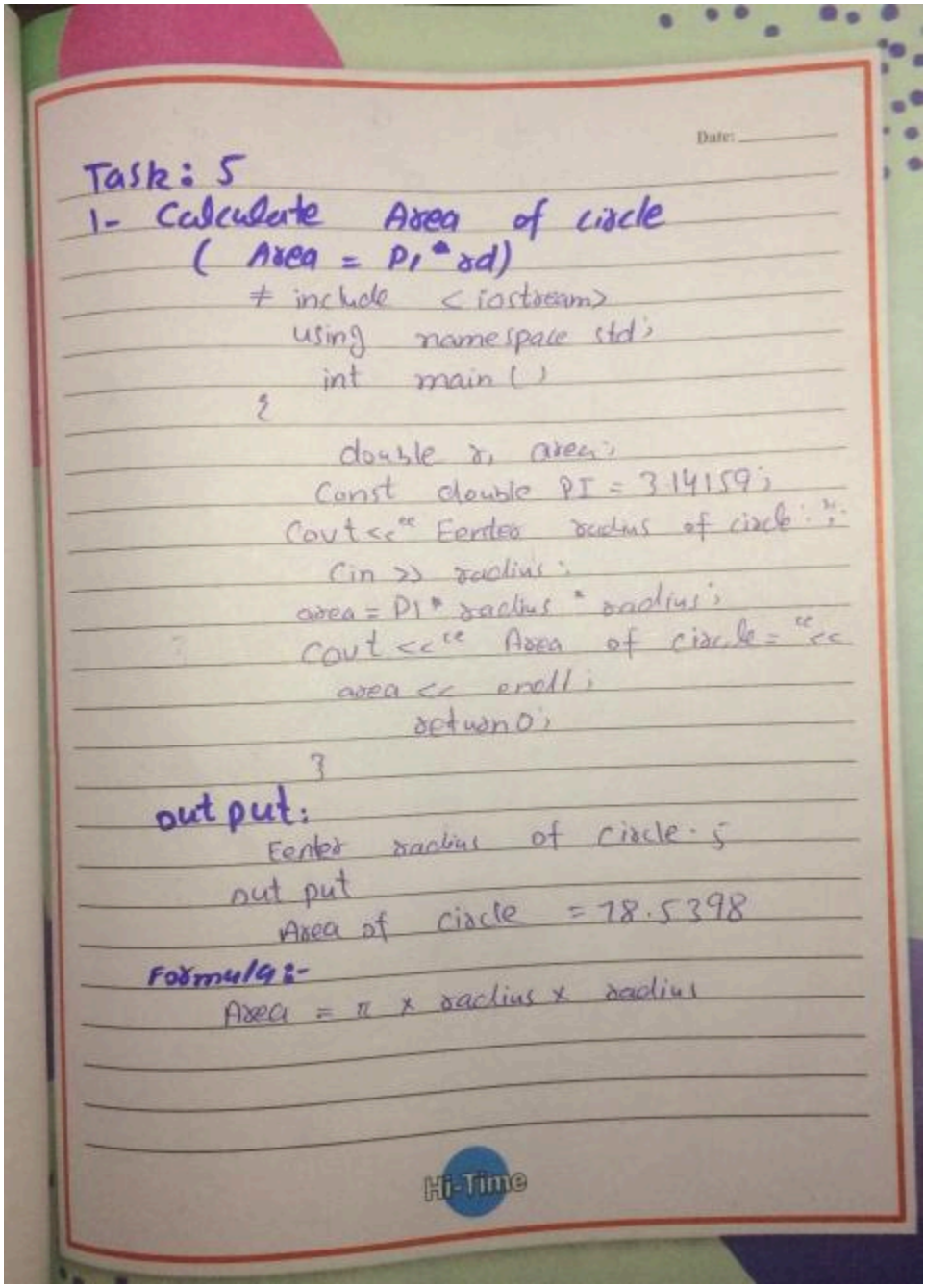
Task 5

```cpp
#include<iostream>
using namespace std;
int main()
{
    // 1. Area of Circle
    double r, areaCircle;
    const double PI = 3.1416;
    cout << "Enter radius of circle: ";
    cin >> r;
    areaCircle = PI * r * r;
    cout << "Area of Circle = " << areaCircle << endl;

    // 2. Area of Rectangle
    double length, width, areaRect;
    cout << "Enter length and width of rectangle: ";
    cin >> length >> width;
    areaRect = length * width;
    cout << "Area of Rectangle = " << areaRect << endl;

    return 0;
}
```

```
Enter radius of circle:
```



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Task: 5

1- Calculate Area of circle
(Area = PI * rd)

```
#include <iostream>
using namespace std;
int main()
{
    double r, area;
    const double PI = 3.14159;
    cout<<" Enter radius of circle: ";
    cin >> radius;
    area = PI * radius * radius;
    cout <<" Area of circle = "<<
    area << endl;
    return 0;
}
```

output:

Enter radius of circle: 5

out put

Area of circle = 78.5398

Formula:-

Area = π x radius x radius

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2- Area of a Rectangle

```
#include <iostream>
using namespace std;
int main()
{
    double length, width, area;
    cout << "Enter length: ";
    cin >> length;
    cout << "Enter width: ";
    cin >> width;
    area = length x width;
    cout << "Area of Rectangle =
    " << area << endl;
        return 0;
    }
```

output:

Enter length: 10
Enter width: 5

output
Area of Rectangle = 50

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3- Area of a triangle

```
#include <iostream>
using namespace std;
int main()
double base, height, area;
Cout << " Enter base: ";
Cin >> base;
Cout << " Enter height: ";
Cin >> height;
area = base * Height x 1/2
Cout << " Area of triangle= " << area << endl;
return 0;
}
```

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Task 6

```cpp
#include<iostream>
using namespace std;
int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal, salesTax, total;
    const double TAX_RATE = 0.06; // 6%

    subtotal = item1 + item2 + item3 +
item4 + item5;
    salesTax = subtotal * TAX_RATE;
    total = subtotal + salesTax;
    cout << "Price of item 1 = $" << item1 <<
endl;
    cout << "Price of item 2 = $" << item2 <<
endl;
    cout << "Price of item 3 = $" << item3 <<
endl;
    cout << "Price of item 4 = $" << item4 <<
endl;
    cout << "Price of item 5 = $" << item5 <<
endl;
    cout << "----------" << endl;
    cout << "Subtotal = $" << subtotal <<
```

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
-------------------------
Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575

[Program finished]
```

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Task:6

Total purchase A customer in a store is purchasing five items.

```
#include <iostream>
using namespace std;
int main()
{
double price of item1 = 12.95;
double Price of item2 = 24.95;
double Price of item3 = 6.95;
double Price of item4 = 14.95;
double Price of item5 = 3.95;
double total = price of item1 + Price of item2
+ Price of item3 + price of item4 + price of item5;
Cout << " total: " << total;
return 0;
}
```

out put

total = 6375

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Home Task 1

```cpp
#include<iostream>
using namespace std;
int main()
{
    int var1, var2, temp;
    cout << "Enter var1 and var2: ";
    cin >> var1 >> var2;

    cout << "Before Swap: var1=" << var1 << " var2=" << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After Swap: var1=" << var1 << " var2=" << var2 << endl;
    return 0;
}
```

```
Enter var1 and var2:
```

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Home task:-

Task-1

```
#include <iostream>
using namespace std;
int main()
{
int var1, var2, tem;
Cout<<" Enter value of var1:";
  Cin >> var1;
Cout << " Enter value of var2:";
  Cin >> var2;
end1;
  Cout<<" var1= " << var1 <<
end1; // Swap the value
  tem = var1;
  var1 = var2;
  var2 = tem;
Cout<<" In After Swapping:" << end1;
Cout<<" var1 = << var1 << end1;
Cout << " var2 = << var2 << end1;
  return 0;
}
```

output:- If you enter value of var1=10 var2=20

Before Swapping
var1= 10
var2 =20

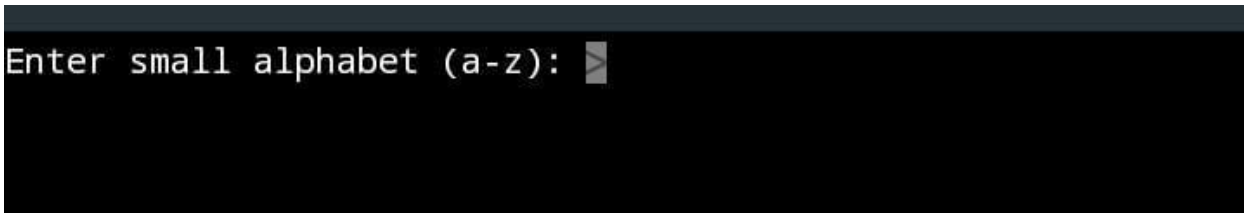
After Swapping
var1 = 20
var2 = 10

Home Task 2

```cpp
#include<iostream>
using namespace std;
int main()
{
    char small, capital;
    cout << "Enter small alphabet (a-z): ";
    cin >> small;

    capital = small - 32; // ASCII difference between a and A is 32

    cout << "Upper case = " << capital;
    return 0;
}
```



Task-2

Date:

```
#include <iostream>
using namespace std;
int main() {
char letter;
Cout << " Enter a small letter: ";
Cin >> letter;
letter = letter - 32;
Cout << " uppercase letter = " << letter
endl;
return 0;
}
```

Out put

Enter a small letter : a

uppercase letter : A

for example :

b → B c → C z → Z

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